



UNIVERSITY OF WOLLONGONG AUSTRALIA

Assignment 3 - Final Report

Setting the environment

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DECLARATION No part of this Assignment has been copied from anyone else, and I have not lent any part of it to anyone else. No part of this assignment has been written by generative AI.



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May 24, 2025

Date

This report was made using R Markdown as a project linking to my Git account

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1 Abstract

- For now, just a heading for this section

2 Glossary

Acronym	Definition
DUMMY	Testing to see if two rows ccan be printed
NOC	National Olympic Committee

3 Introduction

- For now, just a heading for this section

4 Background

I have been looking into some articles regarding the statistics around reducing the number of athletes that a country can send to the Olympics to procure the maximum medals. The following three articles are of interest, and depending on the information, I can extract from my dataset. One of these avenues will become the focus element for this report.

(Haut, Prohl, and Emrich 2014) examines the statistics of the Olympics but also investigates the data from the perspective of increasing funds to rural areas to enhance performance and ultimately achieve more winners. At this stage I don't think that this report will be useful, however there is some regional information in my data set that I can use to plot the locations of the athletes on a world map.

(Condon, Golden, and Wasil 1999) Uses neural networks to produce three models that look at winners from a country perspective using data up until 1996. If I can get the same results, my data set continues to 2016, so this avenue would allow me to extend their findings on the years post 1996 to 2016.

(Heazlewood 2006) Looks into creating models to predict the optimal athlete numbers for all swimming events. This article was able to make some of these predictions, but improvements are needed to apply these results to athletics and swimming across the various distances of the races. These models are applied to results from 2004 and earlier. This article is of particular interest because of the break down of athletes into sections, mainly swimming. It could be useful to attempt to extend their results to apply it to other sports, such as athletics in summer or cross country skiing in winter.

Some of the terminology that I have come across is NOC or National Olympic Committee. Each country has a member that is on the committee and they are responsible for that country. Every athlete must compete under their NOC. I can use this NOC to get a general location of the athletes home town when comparing countries for athlete levels.

5 Research questions and aims of the project

As this is in the early stages of investigation, I still have too many questions to look into.

I can look at athletes, per winter or summer games, per country that the athlete competes at, or per NOC. I can even split the data into per sport. At the end of the day I want to answer, what is the minimal number of athletes that my country should send to the games to receive the most medals based on previous summer and winter Olympic data.

My plan moving forward is to play with the data set some more to see if I can marry the results of prior written research so I can stand on the giants of the past to push the research in this area forward in the future.

6 Rationale

It is no real stretch to underestimate the importance of pride that can come from winning many medals at an Olympic games. From the eyes of the country the cost to participate can be exorbitant to send one athlete, let alone an entire team of athletes. The rationale for this project is to maximize the number of medals that a country can

win, whilst reducing the costs of sending athletes to perform on this stage. I am looking for the optimal number of competing athletes to maximize the gold. In order to look into this problem, we can use past results in order to predict the future. I am unsure at this stage if we can look at this in the scope of the entire country or if we can reduce this to certain sporting events, such as swimming or track and field teams.

7 Data Description

The (Bansal 2021) data set called “Olympics_” was compiled by “Harsh Bansal” and was last updated 4 years ago. The dataset was uploaded and sourced from Kaggle (Keating et al. 2025). According to the site, there is only one owner with no DOI Citation, provenance or license. The restriction on the data is placed on it by Kaggle by way of citation of the owner “Harsh Bansal”. I am using this data at my own risk as it has not been authenticated or carefully curated.

The dataset contains 4 files, “athlete_events_data_dictionary.csv” contains 15 observations of 2 variables, “country_definitions.csv” contains 230 observations of 3 variables, “country_definitions_data_dictionary.csv” contains 3 observations of 2 variables, and “athlete_events.csv” containing 271,116 observations of 15 variables.

The “athlete_events.csv” file contains all athlete information of all the Olympic games dating from 1896 summer games and 1924 winter games up to and including the 2016 summer Olympic games. The following table outlines the variables contained within the set.

Field	Description
ID	Unique number for each athlete
Name	Athlete’s name
Sex	Male (M) or Female (F)
Age	Integer
Height	In centimeters
Weight	In kilograms
Team	Team name
NOC	National Olympic Committee 3-letter code
Games	Year and season
Year	Integer
Season	Summer or Winter
City	Host city
Sport	Sport
Event	Event
Medal	Gold, Silver, Bronze, or NA

The research will center around the NOC, Season, Year, Sport and Medals. I am trying to work out if sending more athletes equates to more medals. If I can isolate what year there is differences in the medal tally for some countries we can pinpoint what if anything was occurring in that country at the time. This may give insight to the specific number of athletes required to send to the games to capitalize on medals.

In the athletes table there is a field called NOC which is the National Olympic City code that links to the country definitions that will allow for better groupings of data by country when linked to each other.

To represent the data on a world map I needed to obtain another data set of data that contained the longitude and latitude. This data set was obtained from (Ferlet 2021) and contained one file, “country-coord.csv”. It was sourced from Github and returned to Tadas Tamošauskas who provided the data set for cleaning. The set is experimental, but provides enough information to roughly plot the Olympic data on a map. It contains six variables. The three that are needed to plot the data points are Alpha-3 code that allows you to link to the NOC codes, latitude (average), and longitude (average). With this data we can obtain an image of what countries have competed in an Olympic games as seen in Figure 1

All Countries that have competed at an olympic games

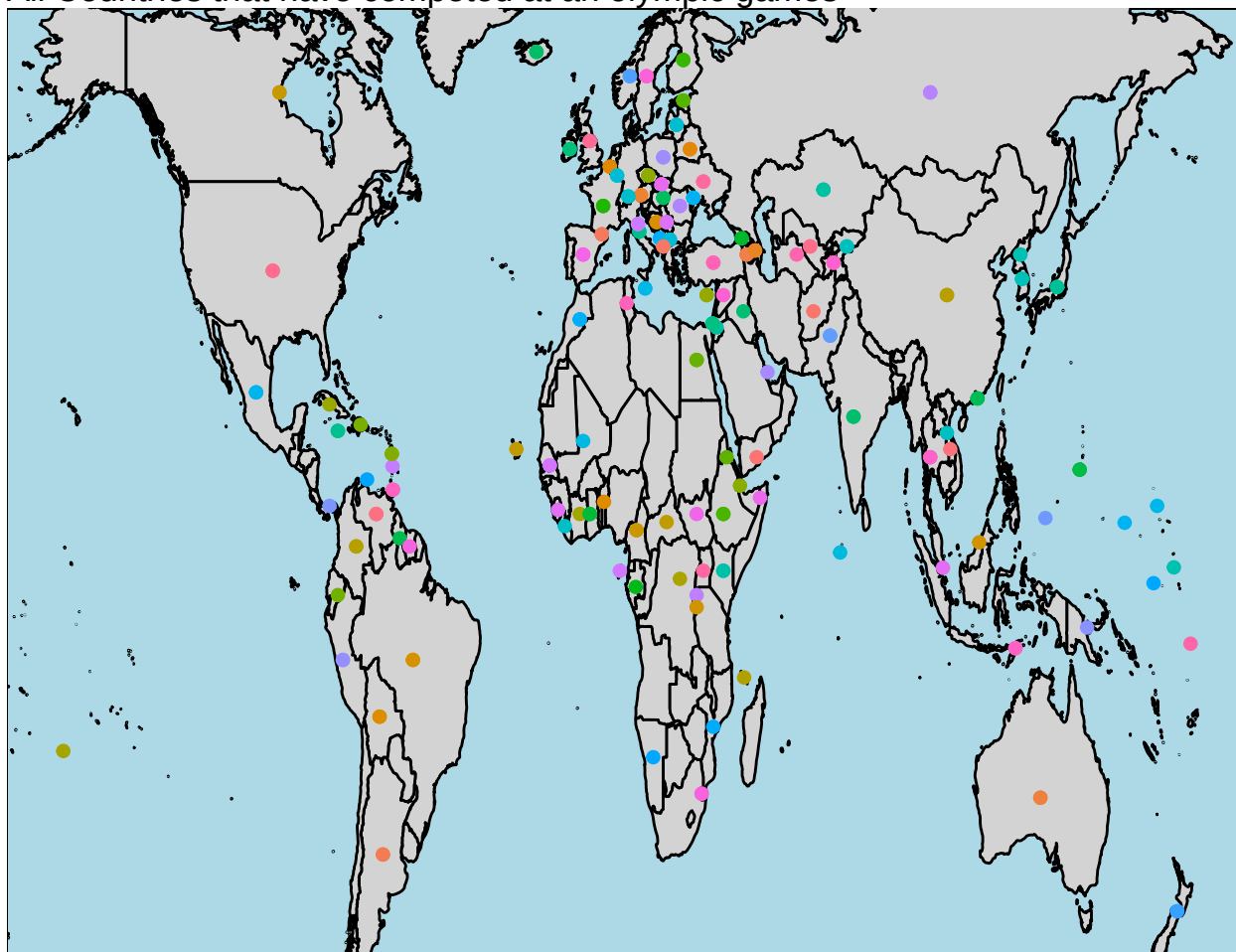


Figure 1: Mapping the NOC codes to longitude and latitude points on the world map

8 Exploratory data analysis

When loading in the data I have determined from the events table, the variables Sex, NOC, Season, and medals are all factor variables and needed to be coded this way in R.

Table 3

Table 4

Table 3: Number of medal winners in 1896 versus 2016 Summer Olympic Games

Sport	Medals_Given	Sport	Medals_Given
Athletics	37	Archery	24
Cycling	16	Athletics	192
Fencing	9	Badminton	24
Gymnastics	37	Basketball	72
Shooting	15	Beach Volleyball	12
Swimming	10	Boxing	51
Tennis	10	Canoeing	82
Weightlifting	6	Cycling	84
Wrestling	3	Diving	36
		Equestrianism	45
		Fencing	65
		Football	106
		Golf	6
		Gymnastics	66
		Handball	89
		Hockey	99
		Judo	56
		Modern Pentathlon	6
		Rhythmic Gymnastics	18
		Rowing	144
		Rugby Sevens	74
		Sailing	45
		Shooting	45
		Swimming	191
		Synchronized Swimming	32
		Table Tennis	24
		Taekwondo	32
		Tennis	24
		Trampolining	6
		Triathlon	6
		Volleyball	72
		Water Polo	78
		Weightlifting	45
		Wrestling	72

Table 4: Winter Sport Total Medals won

Sport	Medals_Given	Sport	Medals_Given
Alpinism	21	Alpine Skiing	31
Bobsleigh	13	Biathlon	60
Cross Country Skiing	6	Bobsleigh	24
Curling	16	Cross Country Skiing	60
Figure Skating	12	Curling	24
Ice Hockey	28	Figure Skating	45
Military Ski Patrol	12	Freestyle Skiing	30
Nordic Combined	3	Ice Hockey	130
Ski Jumping	3	Luge	24
Speed Skating	16	Nordic Combined	18
		Short Track Speed Skating	43
		Skeleton	6
		Ski Jumping	21
		Snowboarding	30
		Speed Skating	51

9 Conclusion / Discussion

Table 5: Medal allocation for 1896 versus 2016 Summer Olympic Games

Sport	Gold	Silver	Bronze	Total	Sport	Gold	Silver	Bronze	Total
Athletics	12	13	12	37	Archery	8	8	8	24
Cycling	6	6	4	16	Athletics	66	64	62	192
Fencing	3	3	3	9	Badminton	8	8	8	24
Gymnastics	26	6	5	37	Basketball	24	24	24	72
Shooting	5	5	5	15	Beach Volleyball	4	4	4	12
Swimming	4	4	2	10	Boxing	13	12	26	51
Tennis	3	3	4	10	Canoeing	27	27	28	82
Weightlifting	2	2	2	6	Cycling	27	28	29	84
Wrestling	1	1	1	3	Diving	12	12	12	36
					Equestrianism	15	15	15	45
					Fencing	21	22	22	65
					Football	35	35	36	106
					Golf	2	2	2	6
					Gymnastics	22	22	22	66
					Handball	29	31	29	89
					Hockey	34	32	33	99
					Judo	14	14	28	56
					Modern Pentathlon	2	2	2	6
					Rhythmic Gymnastics	6	6	6	18
					Rowing	48	48	48	144
					Rugby Sevens	25	24	25	74
					Sailing	15	15	15	45
					Shooting	15	15	15	45
					Swimming	71	63	57	191
					Synchronized Swimming	11	11	10	32
					Table Tennis	8	8	8	24
					Taekwondo	8	8	16	32
					Tennis	8	8	8	24
					Trampolining	2	2	2	6
					Triathlon	2	2	2	6
					Volleyball	24	24	24	72
					Water Polo	26	26	26	78
					Weightlifting	15	15	15	45
					Wrestling	18	18	36	72

Table 6: Medal allocation for 1924 versus 2014 Winter Olympic Games

Sport	Gold	Silver	Bronze	Total	Sport	Gold	Silver	Bronze	Total
Alpinism	21			21	Alpine Skiing	11	9	11	31
Bobsleigh	4	4	5	13	Biathlon	20	20	20	60
Cross Country Skiing	2	2	2	6	Bobsleigh	8	8	8	24
Curling	4	8	4	16	Cross Country Skiing	20	20	20	60
Figure Skating	4	4	4	12	Curling	8	8	8	24
Ice Hockey	9	9	10	28	Figure Skating	16	15	14	45
Military Ski Patrol	4	4	4	12	Freestyle Skiing	10	10	10	30
Nordic Combined	1	1	1	3	Ice Hockey	44	43	43	130
Ski Jumping	1	1	1	3	Luge	8	8	8	24
Speed Skating	5	5	6	16	Nordic Combined	6	6	6	18
					Short Track Speed Skating	15	14	14	43
					Skeleton	2	2	2	6
					Ski Jumping	7	7	7	21
					Snowboarding	10	10	10	30
					Speed Skating	17	17	17	51

Table 7: Medal allocation for 1896 versus 2016 Summer Olympic Games by Team

Sport	Gold	Silver	Bronze	Total	Sport	Gold	Silver	Bronze	Total
Greece	10	18	16	44	United States	52	137	67	256
Germany	24	2	5	31	Germany	43	47	67	157
United States	11	2	7	20	Great Britain	55	64	26	145
France	5	2	4	11	Russia	28	50	35	113
Great Britain	2	2	3	7	China	30	44	35	109
Denmark	1	3	2	6	France	55	20	21	96
Hungary	2	3	1	6	Australia	34	23	25	82
Austria	2	2	1	5	Italy	38	8	24	70
Switzerland	1		2	3	Canada	4	4	61	69
Australia	2			2	Japan	13	17	34	64

#Session Information

sessionInfo()

```
## R version 4.5.0 (2025-04-11 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=English_United States.utf8
## [2] LC_CTYPE=English_United States.utf8
## [3] LC_MONETARY=English_United States.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.utf8
##
## time zone: Australia/Sydney
## tzcode source: internal
##
## attached base packages:
```

```
## [1] stats      graphics  grDevices utils      datasets  methods   base
##
## other attached packages:
## [1] magrittr_2.0.3  maps_3.4.2.1    sjmisc_2.8.10    kableExtra_1.4.0
## [5] sf_1.0-21        lmerTest_3.1-3  lme4_1.1-37      Matrix_1.7-3
## [9] lubridate_1.9.4 forcats_1.0.0    stringr_1.5.1     purrr_1.0.4
## [13] readr_2.1.5      tibble_3.2.1    tidyverse_2.0.0   ggplot2_3.5.2
## [17] dplyr_1.1.4      tidyr_1.3.1     tinytex_0.57      knitr_1.50
##
## loaded via a namespace (and not attached):
## [1] gtable_0.3.6      xfun_0.52        insight_1.3.0
## [4] lattice_0.22-6    tzdb_0.5.0       numDeriv_2016.8-1.1
## [7] vctrs_0.6.5       tools_4.5.0      Rdpack_2.6.4
## [10] generics_0.1.4    proxy_0.4-27     pkgconfig_2.0.3
## [13] KernSmooth_2.23-26 RColorBrewer_1.1-3 lifecycle_1.0.4
## [16] compiler_4.5.0    farver_2.1.2     textshaping_1.0.1
## [19] htmltools_0.5.8.1 class_7.3-23      yaml_2.3.10
## [22] pillar_1.10.2     nloptr_2.2.1     MASS_7.3-65
## [25] classInt_0.4-11   reformulas_0.4.1 boot_1.3-31
## [28] nlme_3.1-168      sjlabelled_1.2.0 tidyselect_1.2.1
## [31] digest_0.6.37     stringi_1.8.7    bookdown_0.43
## [34] labeling_0.4.3     splines_4.5.0    rprojroot_2.0.4
## [37] fastmap_1.2.0     grid_4.5.0       cli_3.6.5
## [40] e1071_1.7-16      withr_3.0.2      scales_1.4.0
## [43] timechange_0.3.0  rmarkdown_2.29   hms_1.1.3
## [46] evaluate_1.0.3    rbibutils_2.3    viridisLite_0.4.2
## [49] rlang_1.1.6       Rcpp_1.0.14      glue_1.8.0
## [52] DBI_1.2.3         xml2_1.3.8       svglite_2.2.1
## [55] rstudioapi_0.17.1 minqa_1.2.8      R6_2.6.1
## [58] systemfonts_1.2.3 units_0.8-7
```

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